

```

159 .
160 .     count if billetera == 1
161 .     local n_usu = r(N)
162 .     count if billetera == 0
163 .     local n_nou = r(N)
164 .     di "N                                     | `n_usu' | `n_nou' |"
165 .     di "=" * 75
166 .
167 .     di ""
168 .     di "PROBABILIDADES CONDICIONALES:"
169 .     foreach bill in 1 0 {
170 .         2.     foreach inf in 1 0 {
171 .             3.         count if billetera == `bill' & informal == `inf'
172 .             4.         local n = r(N)
173 .             5.         sum nuevocredito if billetera == `bill' & informal == `inf'
174 .             6.         local p = r(mean)
175 .             7.         di " Informales=`inf', Billetera=`bill': %9.2f`=100*`p'`% (N=`n')"
176 .             8.         }
177 .         9.     }
178 . }

179 .
180 . end of do-file

181 . clear

182 . import delimited "C:\Users\Alvaro Chicchón\Downloads\BASE_REGRESIONES.csv"
183 . (encoding automatically selected: UTF-8)
184 . (23 vars, 6,361 obs)

185 . clear

186 . import delimited "C:\Users\Alvaro Chicchón\OneDrive - Universidad del Pacífico\UP\Ciclo 2026-II\Investiga
187 . > ción Económica 2\BASE_REGRESIONES.csv"
188 . (encoding automatically selected: UTF-8)
189 . (18 vars, 6,361 obs)

190 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000001.tmp"

191 .
192 . * =====
193 . * A.1. DISEÑO MUESTRAL
194 . * =====
195 .
196 . * Declarar el diseño complejo de la ENAHO
197 .
198 . svyset conglomerado [pw=factor_expansion], strata(estrato)

199 .
200 . Sampling weights: factor_expansion
201 .                   VCE: linearized
202 .                   Single unit: missing
203 .                   Strata 1: estrato
204 . Sampling unit 1: conglomerado
205 .                   FPC 1: <zero>

```

```

185 .
186 . * Para las regresiones se utilizará la alternativa:
187 .
    end of do-file

188 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000m.tmp"

189 .
190 . * =====
191 . * A.2. PREPARACIÓN DE VARIABLES
192 . * =====
193 .
194 . * -----
195 . * Mujer = 1 si la persona es mujer
196 . * -----
197 .
198 . capture drop mujer

199 . gen byte mujer = (sexo == 2) if !missing(sexo)

200 .
201 .
202 . * -----
203 . * Urbano = 1 si el estrato corresponde a zona urbana
204 . * -----
205 .
206 . capture drop urbano

207 . gen byte urbano = (estrato <= 6) if !missing(estrato)

208 .
    end of do-file

209 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000n.tmp"

210 . * -----
211 . * Edad al cuadrado
212 . * -----
213 .
214 . capture drop edad_cuadrado

215 . gen edad_cuadrado = edad^2

216 .
217 .
218 . * -----
219 . * Dummies de dominio
220 . * -----
221 .
222 . capture drop dom1-dom8

223 . tab dominio, gen(dom)

```

dominio	Freq.	Percent	Cum.
1	792	12.45	12.45
2	611	9.61	22.06
3	317	4.98	27.04
4	480	7.55	34.59
5	1,334	20.97	55.56
6	876	13.77	69.33
7	1,455	22.87	92.20
8	496	7.80	100.00
Total	6,361	100.00	

```

224 .
225 .
    end of do-file

226 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000o.tmp"

227 . *      trabajador_informal
228 . * -----
229 .
230 . global X "edad edad_cuadrado mujer nivel_educativo urbano miembros_hogar dom2-dom8"

231 .
232 .
    end of do-file

233 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000p.tmp"

234 .
235 . * =====
236 . * A.2. DESCRIPTIVOS
237 . * =====
238 .
239 . * -----
240 . * Comparación de medias entre usuarios y no usuarios
241 . * de billetera digital
242 . * -----
243 .
244 . table () (usa_billetera), ///
    > stat(mean nuevo_credito_formal trabajador_informal edad ///
    >      mujer nivel_educativo urbano miembros_hogar) ///
    > stat(sd nuevo_credito_formal trabajador_informal edad ///
    >      mujer nivel_educativo urbano miembros_hogar)

```

	usa_billetera		
	0	1	Total
Mean			
nuevo_credito_formal	.0746948	.2041284	.0924383
trabajador_informal	.828384	.5527523	.790599
edad	54.43141	42.84404	52.84295
mujer	.3016943	.3887615	.3136299
nivel_educativo	4.838586	6.864679	5.116334
urbano	.5583895	.9025229	.6055652
miembros_hogar	3.326289	3.473624	3.346486
Standard deviation			
nuevo_credito_formal	.2629223	.4032946	.2896664
trabajador_informal	.3770807	.4974948	.4069131
edad	14.28343	11.69193	14.51382
mujer	.4590351	.4877487	.4640044
nivel_educativo	3.029525	1.954133	2.988024
urbano	.4966242	.2967765	.4887674
miembros_hogar	1.829196	1.530926	1.791855

```

245 .
    end of do-file

246 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000q.tmp"

247 .
248 . * -----
249 . * Diferencia de medias de edad
250 . * -----
251 .
252 . ttest edad, by(usa_billetera)

Two-sample t test with equal variances



| Group    | Obs   | Mean     | Std. err. | Std. dev. | [95% conf. interval] |          |
|----------|-------|----------|-----------|-----------|----------------------|----------|
| 0        | 5,489 | 54.43141 | .1927907  | 14.28343  | 54.05346             | 54.80935 |
| 1        | 872   | 42.84404 | .3959387  | 11.69193  | 42.06693             | 43.62114 |
| Combined | 6,361 | 52.84295 | .1819781  | 14.51382  | 52.48621             | 53.19969 |
| diff     |       | 11.58737 | .5088012  |           | 10.58995             | 12.58479 |



diff = mean(0) - mean(1)
H0: diff = 0
t = 22.7739
Degrees of freedom = 6359

Ha: diff < 0
Pr(T < t) = 1.0000

Ha: diff != 0
Pr(|T| > |t|) = 0.0000

Ha: diff > 0
Pr(T > t) = 0.0000

253 .
254 .
255 . * -----
256 .
    end of do-file

257 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000r.tmp"

258 . * Diferencias estandarizadas
259 . * -----
260 .
261 . * Si no está instalado:
262 . * ssc install stddiff
263 .
264 . stddiff edad nivel_educativo, by(usa_billetera)
    command stddiff is unrecognized
    r(199);

    end of do-file

    r(199);

265 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000s.tmp"

266 . ssc install stddiff
    checking stddiff consistency and verifying not already installed...
    installing into C:\Users\Alvaro Chicchón\ado\plus\...
    installation complete.

```

```

267 .
    end of do-file

268 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000t.tmp"

269 . stddiff edad nivel_educativo, by(usa_billetera)

```

	usa_billetera=0		usa_billetera=1		
	Mean or N	SD or (%)	Mean or N	SD or (%)	Std Diff
edad	54.43	14.283	42.84	11.692	0.88777
nivel educ~o	4.839	3.0295	6.865	1.9541	-0.79480

```

270 .
    end of do-file

271 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000u.tmp"

272 . * -----
273 .
274 . table trabajador_informal usa_billetera, ///
>     stat(mean nuevo_credito_formal) ///
>     stat(freq)

```

	usa_billetera		
	0	1	Total
trabajador_informal			
0			
Mean	.163482	.274359	.1959459
Frequency	942	390	1,332
1			
Mean	.0563009	.1473029	.0650229
Frequency	4,547	482	5,029
Total			
Mean	.0746948	.2041284	.0924383
Frequency	5,489	872	6,361

```

275 .
    end of do-file

276 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000v.tmp"

277 .
278 . probit nuevo_credito_formal ///
>     i.usa_billetera##i.trabajador_informal ///
>     $X ///
>     [pw=factor_expansion], ///
>     vce(cluster conglomerado)

```

```

Iteration 0: Log pseudolikelihood = -583572.88
Iteration 1: Log pseudolikelihood = -544070.75
Iteration 2: Log pseudolikelihood = -542329.24
Iteration 3: Log pseudolikelihood = -542324.87
Iteration 4: Log pseudolikelihood = -542324.87

```

Probit regression

Log pseudolikelihood = **-542324.87**

Number of obs = **6,361**
Wald chi2(16) = **161.45**
Prob > chi2 = **0.0000**
Pseudo R2 = **0.0707**

(Std. err. adjusted for 1,698 clusters in conglomerado)

nuevo_credito_formal	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
1.usa_billetera	.2432255	.1271705	1.91	0.056	-.0060241	.4924751
1.trabajador_informal	-.3570775	.0901764	-3.96	0.000	-.53382	-.180335
usa_billetera#trabajador_informal						
1 1	-.050707	.1681385	-0.30	0.763	-.3802524	.2788384
edad	.0139019	.0176788	0.79	0.432	-.0207478	.0485516
edad_cuadrado	-.0002261	.000175	-1.29	0.196	-.0005691	.0001169
mujer	.0472729	.0713026	0.66	0.507	-.0924777	.1870235
nivel_educativo	.0130367	.0080441	1.62	0.105	-.0027295	.0288029
urbano	.3597179	.0762873	4.72	0.000	.2101975	.5092382
miembros_hogar	-.0155548	.0219121	-0.71	0.478	-.0585017	.0273921
dom2	-.1520756	.1065932	-1.43	0.154	-.3609945	.0568433
dom3	.1331902	.1410471	0.94	0.345	-.143257	.4096373
dom4	-.2548581	.1485554	-1.72	0.086	-.5460213	.0363051
dom5	-.1890553	.1017134	-1.86	0.063	-.38841	.0102993
dom6	-.0003485	.1120586	-0.00	0.998	-.2199794	.2192824
dom7	.0233914	.0941448	0.25	0.804	-.161129	.2079119
dom8	.0312165	.105649	0.30	0.768	-.1758518	.2382848
_cons	-1.46895	.4313298	-3.41	0.001	-2.314341	-.6235594

279 .

end of do-file

280 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000w.tmp"

281 . *

282 .

283 . margins, dydx(usa_billetera) ///

> over(trabajador_informal)

Average marginal effects

Number of obs = 6,361

Model VCE: Robust

Expression: Pr(nuevo_credito_formal), predict()

dy/dx wrt: 1.usa_billetera

Over: trabajador_informal

	Delta-method					
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
0.usa_billetera	(base outcome)					
1.usa_billetera						
trabajador_informal						
0	.0643173	.0348708	1.84	0.065	-.0040281	.1326627
1	.027922	.0193467	1.44	0.149	-.0099968	.0658409

Note: dy/dx for factor levels is the discrete change from the base level.

```

284 .
    end of do-file

285 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_00000x.tmp"

286 .
287 . margins, dydx(usa_billetera) ///
    >       over(trabajador_informal) ///
    >       pwcompare(effects)

```

Pairwise comparisons of average marginal effects

Model VCE: **Robust**

Number of obs = **6,361**

Expression: **Pr(nuevo_credito_formal), predict()**

dy/dx wrt: **1.usa_billetera**

Over: **trabajador_informal**

	Contrast	Delta-method dy/dx	std. err.	Unadjusted z	P> z	Unadjusted [95% conf. interval]
0.usa_billetera	(base outcome)					
1.usa_billetera trabajador_informal 1 vs 0		-.0363953	.0382195	-0.95	0.341	-.1113042 .0385136

Note: dy/dx for factor levels is the discrete change from the base level.

```

288 .
289 .
    end of do-file

290 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000010.tmp"

291 .
292 . margins trabajador_informal#usa_billetera

```

Predictive margins

Number of obs = **6,361**

Model VCE: **Robust**

Expression: **Pr(nuevo_credito_formal), predict()**

	Delta-method Margin	std. err.	z	P> z	[95% conf. interval]	
trabajador_informal#usa_billetera						
0 0	.1332096	.0150574	8.85	0.000	.1036976	.1627215
0 1	.1903172	.0276636	6.88	0.000	.1360976	.2445368
1 0	.0726667	.0068106	10.67	0.000	.0593183	.0860152
1 1	.1020029	.0180027	5.67	0.000	.0667182	.1372876

```

293 .
    end of do-file

294 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000011.tmp"

295 . * -----
296 . * Figura 1
297 . * -----
298 .
299 . marginsplot, ///
    >   recast(scatter) ///
    >   ciopts(recast(rcap))

Variables that uniquely identify margins: trabajador_informal usa_billetera

```

```

300 .
301 .
    end of do-file

302 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000012.tmp"

```

```

303 .
304 . margins, dydx(usa_billetera) ///
    >   over(trabajador_informal) post

```

Average marginal effects Number of obs = **6,361**
 Model VCE: **Robust**

Expression: **Pr(nuevo_credito_formal), predict()**
 dy/dx wrt: **1.usa_billetera**
 Over: **trabajador_informal**

	Delta-method					
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
0.usa_billetera	(base outcome)					
1.usa_billetera						
trabajador_informal						
0	.0643173	.0348708	1.84	0.065	-.0040281	.1326627
1	.027922	.0193467	1.44	0.149	-.0099968	.0658409

Note: dy/dx for factor levels is the discrete change from the base level.

```

305 .
306 . test _b[1.usa_billetera:0.trabajador_informal] = 0.02

    ( 1) [1.usa_billetera]0bn.trabajador_informal = .02

            chi2( 1) = 1.62
            Prob > chi2 = 0.2038

```

```

307 .

```

308 . test _b[1.usa_billetera:0.trabajador_informal] = -0.02

(1) [1.usa_billetera]0bn.trabajador_informal = -.02

chi2(1) = 5.85
Prob > chi2 = 0.0156

309 .
end of do-file

310 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000013.tmp"

311 . * -----

312 .

313 . reg nuevo_credito_formal ///

> i.usa_billetera##i.trabajador_informal ///

> \$X ///

> [pw=factor_expansion], ///

> vce(cluster conglomerado)

(sum of wgt is 1,833,722.5644668)

Linear regression

Number of obs = 6,361
F(16, 1697) = 9.77
Prob > F = 0.0000
R-squared = 0.0448
Root MSE = .28952

(Std. err. adjusted for 1,698 clusters in conglomerado)

nuevo_credito_formal	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.usa_billetera	.0786991	.0362606	2.17	0.030	.007579	.1498192
1.trabajador_informal	-.0639802	.0182649	-3.50	0.000	-.0998043	-.0281562
usa_billetera#trabajador_informal						
1 1	-.0448404	.0421255	-1.06	0.287	-.1274638	.037783
edad	.0001121	.0026186	0.04	0.966	-.0050239	.005248
edad_cuadrado	-.0000129	.0000232	-0.56	0.578	-.0000584	.0000326
mujer	.0094926	.0119044	0.80	0.425	-.0138563	.0328415
nivel_educativo	.0028249	.002188	1.29	0.197	-.0014666	.0071165
urbano	.0384917	.0093353	4.12	0.000	.0201818	.0568016
miembros_hogar	-.0031053	.0032389	-0.96	0.338	-.0094578	.0032473
dom2	-.0257706	.0183942	-1.40	0.161	-.0618483	.0103071
dom3	.0295252	.0297803	0.99	0.322	-.0288848	.0879351
dom4	-.0250247	.0174855	-1.43	0.153	-.0593202	.0092707
dom5	-.0264227	.0150719	-1.75	0.080	-.0559843	.0031388
dom6	-.0006775	.0188313	-0.04	0.971	-.0376125	.0362575
dom7	.0031275	.0163561	0.19	0.848	-.0289528	.0352079
dom8	.0072659	.0213259	0.34	0.733	-.0345619	.0490936
_cons	.1394034	.0746062	1.87	0.062	-.0069265	.2857332

314 .
end of do-file

315 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000014.tmp"

316 . * Efectos marginales del uso de billetera

317 . margins, dydx(usa_billetera) ///
> over(trabajador_informal)

Average marginal effects
Model VCE: **Robust**

Number of obs = **6,361**

Expression: **Linear prediction, predict()**

dy/dx wrt: **1.usa_billetera**

Over: **trabajador_informal**

	Delta-method		t	P> t	[95% conf. interval]	
	dy/dx	std. err.				
0.usa_billetera	(base outcome)					
1.usa_billetera						
trabajador_informal						
0	.0786991	.0362606	2.17	0.030	.007579	.1498192
1	.0338587	.0248148	1.36	0.173	-.0148122	.0825296

Note: dy/dx for factor levels is the discrete change from the base level.

318 .
end of do-file

319 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000015.tmp"

320 .

321 . probit nuevo_credito_formal ///
> i.usa_billetera##i.trabajador_informal ///
> \$X ///
> [pw=factor_expansion], ///
> vce(cluster conglomerado)

Iteration 0: Log pseudolikelihood = **-583572.88**

Iteration 1: Log pseudolikelihood = **-544070.75**

Iteration 2: Log pseudolikelihood = **-542329.24**

Iteration 3: Log pseudolikelihood = **-542324.87**

Iteration 4: Log pseudolikelihood = **-542324.87**

Probit regression

Number of obs = **6,361**

Wald chi2(16) = **161.45**

Prob > chi2 = **0.0000**

Pseudo R2 = **0.0707**

Log pseudolikelihood = **-542324.87**

(Std. err. adjusted for **1,698** clusters in **conglomerado**)

nuevo_credito_formal	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
1.usa_billetera	.2432255	.1271705	1.91	0.056	-.0060241	.4924751
1.trabajador_informal	-.3570775	.0901764	-3.96	0.000	-.53382	-.180335
usa_billetera#trabajador_informal						
1 1	-.050707	.1681385	-0.30	0.763	-.3802524	.2788384
edad	.0139019	.0176788	0.79	0.432	-.0207478	.0485516
edad_cuadrado	-.0002261	.000175	-1.29	0.196	-.0005691	.0001169
mujer	.0472729	.0713026	0.66	0.507	-.0924777	.1870235

nivel_educativo	.0130367	.0080441	1.62	0.105	-.0027295	.0288029
urbano	.3597179	.0762873	4.72	0.000	.2101975	.5092382
miembros_hogar	-.0155548	.0219121	-0.71	0.478	-.0585017	.0273921
dom2	-.1520756	.1065932	-1.43	0.154	-.3609945	.0568433
dom3	.1331902	.1410471	0.94	0.345	-.143257	.4096373
dom4	-.2548581	.1485554	-1.72	0.086	-.5460213	.0363051
dom5	-.1890553	.1017134	-1.86	0.063	-.38841	.0102993
dom6	-.0003485	.1120586	-0.00	0.998	-.2199794	.2192824
dom7	.0233914	.0941448	0.25	0.804	-.161129	.2079119
dom8	.0312165	.105649	0.30	0.768	-.1758518	.2382848
_cons	-1.46895	.4313298	-3.41	0.001	-2.314341	-.6235594

```

322 .
323 . margins, dydx(usa_billetera) ///
>       over(trabajador_informal)

```

Average marginal effects
Model VCE: **Robust**

Number of obs = **6,361**

Expression: **Pr(nuevo_credito_formal), predict()**
dy/dx wrt: **1.usa_billetera**
Over: **trabajador_informal**

	Delta-method					
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
0.usa_billetera	(base outcome)					
1.usa_billetera						
trabajador_informal						
0	.0643173	.0348708	1.84	0.065	-.0040281	.1326627
1	.027922	.0193467	1.44	0.149	-.0099968	.0658409

Note: dy/dx for factor levels is the discrete change from the base level.

```

324 .
    end of do-file

325 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000016.tmp"

326 . ssc install psacalc
    checking psacalc consistency and verifying not already installed...
    installing into C:\Users\Alvaro Chicchón\ado\plus\...
    installation complete.

327 .
    end of do-file

328 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000017.tmp"

329 . * Especificación completa
330 . reg nuevo_credito_formal ///
>     usa_billetera ///
>     $X

```

Source	SS	df	MS	Number of obs	=	6,361
Model	28.7158117	14	2.0511294	F(14, 6346)	=	25.78
Residual	504.93047	6,346	.07956673	Prob > F	=	0.0000
				R-squared	=	0.0538
				Adj R-squared	=	0.0517
Total	533.646282	6,360	.083906648	Root MSE	=	.28208

nuevo_credito~1	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
usa_billetera	.0801834	.0112888	7.10	0.000	.0580535	.1023133
edad	.0010536	.0015893	0.66	0.507	-.0020619	.0041691
edad_cuadrado	-.000022	.0000146	-1.50	0.132	-.0000506	6.65e-06
mujer	-.001377	.0078747	-0.17	0.861	-.016814	.0140601
nivel_educativo	.009046	.0013001	6.96	0.000	.0064972	.0115947
urbano	.0486653	.0081862	5.94	0.000	.0326176	.064713
miembros_hogar	-.0046435	.0021446	-2.17	0.030	-.0088476	-.0004395
dom2	-.033386	.0152213	-2.19	0.028	-.0632248	-.0035472
dom3	.0354472	.0188424	1.88	0.060	-.0014904	.0723847
dom4	-.0297814	.0167786	-1.77	0.076	-.0626731	.0031103
dom5	-.0267399	.0129287	-2.07	0.039	-.0520846	-.0013953
dom6	-.0146542	.0140846	-1.04	0.298	-.0422648	.0129565
dom7	.0027604	.012662	0.22	0.827	-.0220613	.0275822
dom8	-.025587	.0163274	-1.57	0.117	-.0575942	.0064202
_cons	.044682	.0442145	1.01	0.312	-.0419934	.1313574

```

331 .
332 .
333 . * Definir Rmax como 1.3 veces el R2
334 . local rmax = 1.3 * e(r2)

335 .
336 .
337 . * Cota de Oster
338 . psacalc delta usa_billetera, rmax(`rmax`)

```

—— Bound Estimate ——		
delta	2.06887	
—— Inputs from Regressions ——		
	Coeff.	R-Squared
Uncontrolled	0.12943	0.024
Controlled	0.08018	0.054
—— Other Inputs ——		
R_max	0.070	
Beta	0.000000	
Unr. Controls		

```

339 .
    end of do-file

340 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000018.tmp"

```

```

341 .
342 . probit nuevo_credito_formal ///
>     i.usa_billetera##i.trabajador_informal ///
>     $X, ///
>     vce(cluster conglomerado)

```

```

Iteration 0:  Log pseudolikelihood = -1960.0985
Iteration 1:  Log pseudolikelihood = -1769.3478
Iteration 2:  Log pseudolikelihood = -1757.7438
Iteration 3:  Log pseudolikelihood = -1757.6945
Iteration 4:  Log pseudolikelihood = -1757.6945

```

Probit regression

```

Number of obs = 6,361
Wald chi2(16) = 322.26
Prob > chi2    = 0.0000
Pseudo R2     = 0.1033

```

Log pseudolikelihood = -1757.6945

(Std. err. adjusted for 1,698 clusters in conglomerado)

nuevo_credito_formal	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
1.usa_billetera	.2313145	.0858768	2.69	0.007	.0629991	.3996298
1.trabajador_informal	-.4525842	.0656342	-6.90	0.000	-.5812249	-.3239435
usa_billetera#trabajador_informal						
1 1	.0543929	.1173641	0.46	0.643	-.1756366	.2844224
edad	.0259047	.0112316	2.31	0.021	.0038911	.0479183
edad_cuadrado	-.0003648	.0001115	-3.27	0.001	-.0005833	-.0001463
mujer	-.0064583	.0498142	-0.13	0.897	-.1040923	.0911757
nivel_educativo	.0263361	.010339	2.55	0.011	.006072	.0466001
urbano	.3434213	.0650565	5.28	0.000	.215913	.4709296
miembros_hogar	-.0253132	.0146153	-1.73	0.083	-.0539587	.0033322
dom2	-.2135079	.0985009	-2.17	0.030	-.406566	-.0204497
dom3	.1188581	.1083329	1.10	0.273	-.0934705	.3311866
dom4	-.3046074	.1312347	-2.32	0.020	-.5618227	-.0473921
dom5	-.1995847	.0917379	-2.18	0.030	-.3793877	-.0197817
dom6	-.104418	.0933019	-1.12	0.263	-.2872864	.0784504
dom7	.0069676	.0817475	0.09	0.932	-.1532545	.1671897
dom8	-.1776904	.0984231	-1.81	0.071	-.3705962	.0152155
_cons	-1.570341	.2961829	-5.30	0.000	-2.150848	-.9898328

```

343 .
344 .
345 . * Efectos marginales sin ponderadores
346 . margins, dydx(usa_billetera) ///
>     over(trabajador_informal)

```

Average marginal effects
Model VCE: Robust

Number of obs = 6,361

Expression: Pr(nuevo_credito_formal), predict()
dy/dx wrt: 1.usa_billetera
Over: trabajador_informal

	Delta-method		z	P> z	[95% conf. interval]	
	dy/dx	std. err.				
0.usa_billetera	(base outcome)					
1.usa_billetera						
trabajador_informal						
0	.0638772	.0244014	2.62	0.009	.0160513	.111703
1	.0394373	.013288	2.97	0.003	.0133934	.0654813

Note: dy/dx for factor levels is the discrete change from the base level.

347 .

348 .

end of do-file

349 . do "C:\Users\ALVARO~1\AppData\Local\Temp\STD7140_000019.tmp"

```
350 . probit credito_formal_2025 ///
> i.usa_billetera##i.trabajador_informal ///
> $X ///
> [pw=factor_expansion], ///
> vce(cluster conglomerado)
```

Iteration 0: Log pseudolikelihood = **-583572.88**
Iteration 1: Log pseudolikelihood = **-544070.75**
Iteration 2: Log pseudolikelihood = **-542329.24**
Iteration 3: Log pseudolikelihood = **-542324.87**
Iteration 4: Log pseudolikelihood = **-542324.87**

Probit regression

Number of obs = **6,361**

Wald chi2(16) = **161.45**

Prob > chi2 = **0.0000**

Pseudo R2 = **0.0707**

Log pseudolikelihood = **-542324.87**

(Std. err. adjusted for **1,698** clusters in **conglomerado**)

credito_formal_2025	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
1.usa_billetera	.2432255	.1271705	1.91	0.056	-.0060241	.4924751
1.trabajador_informal	-.3570775	.0901764	-3.96	0.000	-.53382	-.180335
usa_billetera#trabajador_informal						
1 1	-.050707	.1681385	-0.30	0.763	-.3802524	.2788384
edad	.0139019	.0176788	0.79	0.432	-.0207478	.0485516
edad_cuadrado	-.0002261	.000175	-1.29	0.196	-.0005691	.0001169
mujer	.0472729	.0713026	0.66	0.507	-.0924777	.1870235
nivel_educativo	.0130367	.0080441	1.62	0.105	-.0027295	.0288029
urbano	.3597179	.0762873	4.72	0.000	.2101975	.5092382
miembros_hogar	-.0155548	.0219121	-0.71	0.478	-.0585017	.0273921
dom2	-.1520756	.1065932	-1.43	0.154	-.3609945	.0568433
dom3	.1331902	.1410471	0.94	0.345	-.143257	.4096373
dom4	-.2548581	.1485554	-1.72	0.086	-.5460213	.0363051
dom5	-.1890553	.1017134	-1.86	0.063	-.38841	.0102993
dom6	-.0003485	.1120586	-0.00	0.998	-.2199794	.2192824
dom7	.0233914	.0941448	0.25	0.804	-.161129	.2079119
dom8	.0312165	.105649	0.30	0.768	-.1758518	.2382848
_cons	-1.46895	.4313298	-3.41	0.001	-2.314341	-.6235594

```
351 .
352 .
353 . * Efectos marginales
354 . margins, dydx(usa_billetera) ///
> over(trabajador_informal)
```

Average marginal effects
Model VCE: **Robust**

Number of obs = **6,361**

Expression: **Pr(credito_formal_2025), predict()**
dy/dx wrt: **1.usa_billetera**
Over: **trabajador_informal**

	Delta-method		z	P> z	[95% conf. interval]	
	dy/dx	std. err.				
0.usa_billetera	(base outcome)					
1.usa_billetera						
trabajador_informal						
0	.0643173	.0348708	1.84	0.065	-.0040281	.1326627
1	.027922	.0193467	1.44	0.149	-.0099968	.0658409

Note: dy/dx for factor levels is the discrete change from the base level.

```
355 .
end of do-file
```

```
356 .
```